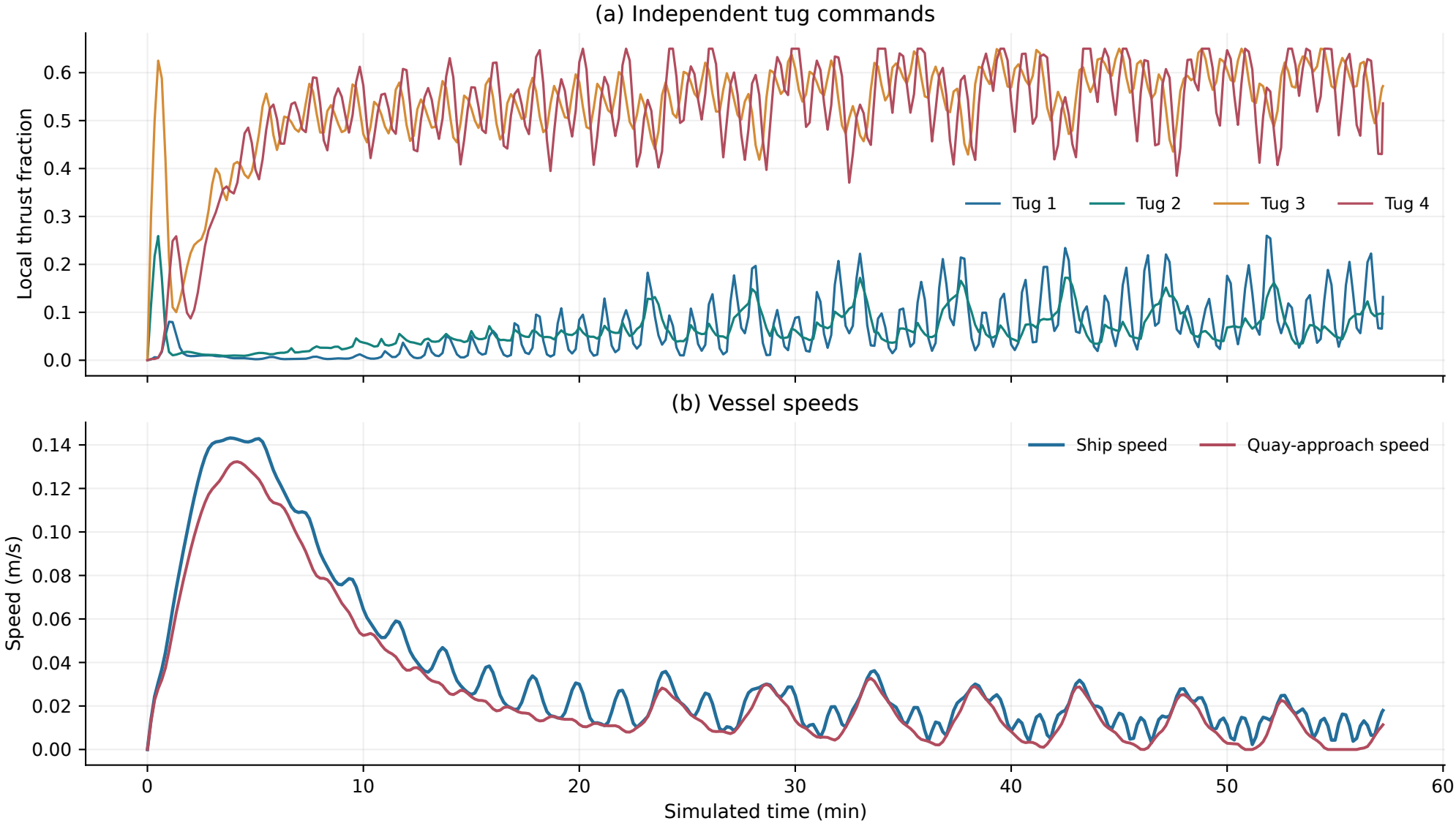
(a) Independent tug commands
Local thrust fraction
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Tug 1
Tug 2
Tug 3
Tug 4
(b) Vessel speeds
Speed (m/s)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Ship speed
Quay-approach speed
0
10
20
30
40
50
60
Simulated time (min)